XVisionNet v3: An Explainable Hybrid Deep Learning Framework for Protocol-Aware Network Intrusion Detection

PHASE I REPORT

Submitted by

NAREAN PRASATH A K
NITESH P K
PRASANNA S
KESAVAN K

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Signature of the HOD with date

Name of the HOD

Professor and Head

Department of Computer Science and Engineering

KPR Institute of Engineering and Technology

Signature of the Supervisor with date

Name of the Supervisor

Academic Designation of Supervisor

ANNEXURE III

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF TABLES	xvi
	LIST OF FIGURES	xviii
	LIST OF SYMBOLS, ABBREVIATIONS	xxvii
1	**INTRODUCTION**	1
	1.1 GENERAL	1
	1.2 NEED FOR THE STUDY	2
	1.3 OBJECTIVES OF THE STUDY	3
2	**REVIEW OF LITERATURE**	4
	2.1 INTRODUCTION	4
	2.2 FRAMEWORK OF LCA	4
	2.2.1 Product Life Cycle	6
	2.2.2 Product System and System Boundary	6
	2.2.3 Functional Unit and Reference Flow	7
	2.2.4 Environmental Burdens	8
	2.2.5 Environmental Impacts	8

CHAPTER 1

INTRODUCTION

1.1 GENERAL

INTRODUCTION

The rapid expansion of interconnected computer networks and Internet-based services has significantly increased the exposure of organizations to cyberattacks and network intrusions. Modern network environments generate enormous volumes of heterogeneous traffic across protocols such as TCP, UDP, HTTP, DNS, and ICMP, making it increasingly difficult for traditional rule-based intrusion detection systems to identify sophisticated and evolving attacks. While conventional Intrusion Detection Systems (IDS) can detect known attack patterns using predefined signatures, they often struggle with zero-day attacks, complex traffic behaviors, and highly imbalanced network datasets. Most importantly, simply classifying network traffic as normal or malicious does not provide security analysts with sufficient information about why a particular connection was identified as an attack. **XVisionNet v3** is designed as an explainable hybrid deep learning framework that combines multiple deep learning techniques with protocol-aware traffic analysis to detect malicious network activities accurately. By incorporating Explainable Artificial Intelligence (XAI), the framework provides meaningful insights into the features and traffic characteristics that influence each prediction, enabling security analysts to understand, validate, and respond to detected threats more effectively.

1.2 NEED FOR THE STUDY

Existing network intrusion detection systems fall short in several important ways when dealing with modern and evolving cyber threats. Traditional Signature-Based IDS rely on predefined attack patterns and are effective mainly against known threats, but they struggle to detect zero-day attacks, polymorphic attacks, and previously unseen network behaviors. Conventional Machine Learning and Deep Learning models can identify complex patterns in network traffic, but many operate as black-box systems, providing limited insight into why a particular connection is classified as malicious. Protocol-Agnostic detection

approaches also treat different types of network traffic uniformly, potentially overlooking protocol-specific characteristics that can provide valuable indicators of an intrusion.

There is a critical need for a system that combines the predictive capabilities of **hybrid deep learning models** with **protocol-aware network traffic analysis**, while also providing transparent and understandable predictions. **XVisionNet v3** addresses this need by integrating multiple deep learning techniques with Explainable Artificial Intelligence (XAI), enabling accurate intrusion detection while identifying the important traffic features and protocol characteristics responsible for each prediction. This combination aims to provide a more **accurate, interpretable, and security-analyst-friendly** approach to modern network intrusion detection.

1.3 OBJECTIVES OF THE STUDY

The primary objectives of this study are: (1) To develop a **protocol-aware network intrusion detection framework** capable of accurately distinguishing normal network traffic from different types of malicious activities across multiple network protocols. (2) To implement a **hybrid deep learning architecture** that combines the strengths of multiple deep learning techniques to learn complex and diverse patterns from network traffic and improve detection performance. (3) To incorporate **Explainable Artificial Intelligence (XAI)** techniques to identify and present the important network features and protocol characteristics influencing each intrusion prediction, reducing the black-box nature of deep learning models. (4) To improve the reliability and practical usability of intrusion detection by providing **accurate, interpretable, and actionable security insights** that can assist network administrators and security analysts in understanding and responding to potential cyber threats.

CHAPTER 2

REVIEW OF LITERATURE

2.1 INTRODUCTION

This chapter reviews existing systems and technologies utilized for **network intrusion detection**, highlighting their strengths and identifying the critical gaps that **XVisionNet v3** aims to address. It covers traditional signature-based intrusion detection systems, machine learning and deep learning-based detection approaches, hybrid deep learning architectures, protocol-aware network traffic analysis, and the implementation of **Explainable Artificial Intelligence (XAI)** for improving the transparency and interpretability of intrusion detection systems.

2.2 EXISTING NETWORK INTRUSION DETECTION SYSTEMS

General intrusion detection systems attempt to identify malicious network activities by analyzing traffic patterns and comparing them against known attack characteristics. For instance, **signature-based Intrusion Detection Systems (IDS)** use predefined rules and attack signatures to detect previously identified threats. However, these systems are largely dependent on regularly updated signatures and offer limited capability in detecting zero-day attacks or previously unseen attack patterns. Similarly, conventional **machine learning-based IDS** can learn patterns from network traffic and improve detection beyond fixed signatures, but their performance may be affected by imbalanced datasets, complex traffic behavior, and insufficient consideration of protocol-specific characteristics.

Deep learning-based systems such as **CNN, RNN, and LSTM-based IDS** can automatically learn complex representations from network traffic and achieve improved detection performance. However, many such models operate as black boxes and provide limited information about why a particular traffic instance is classified as malicious. These limitations highlight the need for a system that combines the predictive capabilities of deep learning with **protocol-aware analysis and explainable AI**, which are key aspects addressed by **XVisionNet v3**.

2.3 MACHINE LEARNING AND DEEP LEARNING-BASED INTRUSION DETECTION SYSTEMS

Several modern intrusion detection systems have adopted Machine Learning (ML) and Deep Learning (DL) techniques to automate the identification of malicious network activities. Machine learning approaches such as Decision Trees, Random Forest, Support Vector Machines, and K-Nearest Neighbors can learn attack patterns from labeled network traffic and classify connections as normal or malicious. However, their effectiveness depends heavily on the quality of extracted features and the availability of representative training data.

2.4 ARTIFICIAL INTELLIGENCE AND EXPLAINABLE INTRUSION DETECTION

Artificial Intelligence and Deep Learning models have demonstrated significant potential in network security by automatically analyzing large volumes of network traffic and identifying complex attack patterns. However, when these models operate without an explainability layer, their predictions are often difficult for security analysts to interpret. A model may classify a network connection as malicious with high confidence, while providing limited information about which traffic characteristics or protocol features influenced the decision.

To address this limitation, recent advancements in Explainable Artificial Intelligence (XAI) techniques provide methods for interpreting machine learning and deep learning predictions by identifying the features that contribute most significantly to a particular classification. Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) can generate human-understandable explanations for individual intrusion predictions. XVisionNet v3 adopts this explainability approach alongside its hybrid deep learning architecture and protocol-aware analysis, enabling security analysts to understand not only whether network traffic is malicious, but also why the system considers it suspicious.

CHAPTER 3

METHODOLOGY AND SYSTEM ARCHITECTURE

3.1 INTRODUCTION

The methodology of **XVisionNet v3** revolves around addressing the limitations of conventional network intrusion detection systems by combining hybrid deep learning, protocol-aware traffic analysis, and Explainable Artificial Intelligence (XAI). The system is designed to analyze network traffic, identify complex patterns associated with malicious activities, and classify connections into normal and attack categories. To improve both detection performance and interpretability, the architecture separates the intrusion detection process from the explanation generation process, allowing the deep learning model to focus on accurate classification while the explainability layer provides meaningful insights into the reasons behind each prediction.

3.2 SYSTEM ARCHITECTURE: HYBRID DETECTION AND EXPLAINABILITY LAYERS

The core design of **XVisionNet v3** operates through two major layers. **Layer 1 (The Detection Layer)** consists of the protocol-aware preprocessing pipeline and hybrid deep learning model. It receives network traffic, extracts relevant flow-based and protocol-specific features, performs preprocessing and feature transformation, and classifies the traffic as normal or malicious. The hybrid architecture combines the strengths of multiple deep learning techniques to capture both spatial and sequential characteristics present in network traffic.

Layer 2 (The Explainability Layer) is activated after the detection model produces its prediction. It uses Explainable Artificial Intelligence techniques such as **SHAP and LIME** to identify the features that contributed significantly to the model's decision. The resulting explanations provide security analysts with interpretable information about the network traffic, such as unusual packet behavior, protocol characteristics, connection patterns, or abnormal feature values. This separation ensures that the prediction process remains focused

on accurate classification while the explanation layer makes the final decision easier to understand and validate.

3.3 THE XVISIONNET V3 HYBRID DEEP LEARNING FRAMEWORK

The **XVisionNet v3** framework uses a hybrid deep learning architecture to learn both feature patterns and sequential relationships in network traffic. This enables effective identification of normal and malicious traffic patterns.

The framework performs preprocessing such as data cleaning, normalization, feature transformation, and protocol encoding. The processed data is then classified by the hybrid model into normal or attack categories, improving detection accuracy and reducing false classifications.

3.4 PROTOCOL-AWARE INTRUSION DETECTION AND EXPLAINABILITY

XVisionNet v3 incorporates protocol-aware analysis to ensure that network traffic is analyzed according to its communication characteristics. Different protocols such as **TCP, UDP, HTTP, DNS, and ICMP** exhibit different traffic behaviors, packet structures, and communication patterns. Considering these protocol-specific characteristics allows the framework to identify abnormal behavior that may otherwise be overlooked by protocol-agnostic detection approaches.

The system extracts and analyzes relevant protocol and flow-level features along with other network traffic characteristics. These features are provided to the hybrid deep learning model for intrusion classification. After a prediction is generated, the explainability layer analyzes the model's decision and identifies the most influential features contributing to the classification. This enables the system to provide not only an intrusion prediction but also an understandable explanation of the factors responsible for that prediction. As a result, **XVisionNet v3** aims to provide a more transparent, reliable, and practical solution for modern network intrusion detection.

CHAPTER 4

IMPLEMENTATION AND PRELIMINARY RESULTS

4.1 DATASET AND TECHNOLOGY STACK

The framework was evaluated using a network intrusion detection dataset containing normal and malicious network traffic samples across multiple attack categories. The implementation uses freely available software and machine learning components. The technology stack includes Python for preprocessing and model development, TensorFlow/Keras for the hybrid deep learning architecture, and Pandas and NumPy for data processing. Scikit-learn is used for feature encoding, normalization, and evaluation, while SHAP and LIME provide explainable predictions and feature-importance analysis.

4.2 INTERNAL CONSISTENCY AND EVALUATION

Preliminary results focused on Internal Consistency Testing to ensure that the system correctly followed its defined preprocessing, classification, and explainability procedures. Testing across the network traffic dataset confirmed that the preprocessing pipeline consistently transformed the input features into the required format for the hybrid deep learning model. The system successfully classified traffic into normal and malicious categories while maintaining consistent prediction behavior across different protocol types. The evaluation also confirmed that the explainability layer identified important features contributing to individual predictions, helping to verify whether the detected network behavior was associated with potential intrusion activity.

4.3 USER INTERFACE AND OUTPUTS

The finalized outputs clearly present the intrusion detection result along with the corresponding explanation. Detected network traffic is classified as normal or malicious, while relevant protocol and traffic features are displayed to support the prediction. The system also provides feature-importance information through the explainability layer, allowing security analysts to understand the factors responsible for a particular classification. Additionally, the interface presents the detected attack category.

CHAPTER 5

CONCLUSIONS AND WORK SCHEDULE

5.1 CONCLUSIONS

XVisionNet v3 successfully demonstrates an explainable hybrid deep learning approach for protocol-aware network intrusion detection. By combining hybrid deep learning with protocol-specific traffic analysis, the framework can identify normal and malicious network activities more effectively. The integration of SHAP and LIME provides meaningful explanations for model predictions, reducing the black-box nature of deep learning. The Phase I implementation demonstrates the feasibility of combining accurate intrusion detection with transparent and understandable security analysis.

5.2 WORK SCHEDULE FOR PHASE II

The following tasks are scheduled for execution during Phase II of the project: (1) Improving the hybrid deep learning model to enhance intrusion detection accuracy and reduce false positives. (2) Expanding the dataset with additional network protocols and diverse attack types. (3) Conducting extensive performance evaluation using accuracy, precision, recall, F1-score, and other relevant metrics. (4) Enhancing the explainability layer to provide clearer feature-importance analysis and more useful insights for security analysts.

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